

Local Minimax Geometry of Uniform Convolution Factors

Mixed collision strata, boundary-localized Gaussian confounding,
and regular variance estimation at positive collisions

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Abstract

A global recovery rate can conceal radically different local statistical geometries. We study a known bounded background convolved with a fixed-capacity list of centered uniform laws and a Gaussian. At a reference factorization, let r half-lengths be zero and let the distinct positive half-lengths have multiplicities $m_1, \dots, m_s$. We prove matching minimax bounds on every sufficiently small fixed neighborhood of this configuration. A positive cluster of multiplicity m_j has optimal half-length rate $n^{-1/(2m_j)}$, whether the Gaussian variance is known or unknown. The zero block, when present, has rate $n^{-1/(4r)}$ for known variance and $n^{-1/(4r+4)}$ for unknown variance. The unknown Gaussian variance itself has rate $n^{-1/(2r+2)}$; in particular, it remains root- n estimable when $r = 0$, even at arbitrarily high positive multiplicities. We also construct an asymptotically normal variance estimator in that case and give its influence polynomial explicitly.

The main algebraic results are a cluster-local shifted-power-sum inequality and a positivity certificate for the missing first power sum. They separate positive collisions from the zero boundary without assuming that the clusters can be observed separately. One global moment estimator adapts to all fixed strata. Exact moment-preserving paths and weighted likelihood expansions give matching lower bounds and explicit Hellinger constants. The results refine, rather than reassert, the repository's existing global Gaussian-confounding rates. All central arguments are supplied as conventional proofs; the finite checks are supplementary, and no Lean verification or worldwide priority claim is made. Eight further research questions identify the remaining multiscale, efficiency, and unsmoothed-boundary problems.

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1 The research question and the contribution boundary

1.1 From a worst-case rate to a local classification

The Fabius–Rvachev branch of ProveIt studies exact uniform convolutions and inverse problems associated with them. Its report *Gaussian Confounding and Sharp Recovery of Uniform Convolution Factors* [1] establishes global half-length rates $n^{-1/(4M)}$ with known Gaussian variance and $n^{-1/(4M+4)}$ with unknown variance, together with variance rate $n^{-1/(2M+2)}$. Here M is the maximum number of uniform factors, with absent factors represented by zero half-lengths. Its companion flat-boundary report extends the global theory to unsmoothed infinite-uniform backgrounds and explicitly leaves mixed collision strata among its further questions [2]. These global statements are prior results, not discoveries claimed in this article.

They leave a concrete refinement: what happens near a factorization containing *both* nonzero collisions and vanishing factors? Does a zero block slow recovery of a distant positive block? Does estimating the Gaussian variance necessarily incur the same singularity as separating repeated positive roots? Can a single estimator attain the appropriate rates without being supplied the correct multiplicity pattern?

We answer these questions for a positive Gaussian smoothing variance. The answer is not obtained by replacing the global capacity by one universal “effective number of factors.” There are several simultaneously relevant exponents. Each positive cluster has its own multiplicity exponent; the zero block has an additional square-root singularity; and the missing variance information adds an extra order *only at zero*. The noise variance depends on the size of the zero block, not on the largest positive multiplicity.

1.2 What is new relative to the inspected sources

The principal extension is the full blockwise statistical classification in Theorem 2.1, including matching lower bounds and a stratum-adaptive estimator. Its algebraic inputs are the localized shifted-moment theorem and the missing-first-sum certificate. The regular variance result and explicit influence polynomial at positive collisions are additional consequences. The all-zero specialization recovers the earlier global exponents.

Cumulants, Newton identities, real-root perturbations, and moment matching are classical tools. The geometry of the Prony map and its collision singularities is studied by Batenkov–Yomdin [3]. Local minimax geometry and the distinction between local-uniform and pointwise rates are important in the finite-mixture work of Heinrich–Kahn [4]; denoised moment fitting and variance estimation appear in Wu–Yang [5]. Those models concern *mixtures*, whereas our model is a *sum of independent uniforms*. Their unknown weights and parameter metrics differ. Their results are context, not substitutes for a proof of the exponents below.

The repository comparison is pinned to

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The accompanying source audit distinguishes the inspected statements from broader, unverified novelty claims. This is a proof-bearing research article, not a claim to have settled an unrelated named conjecture. Its mathematical arguments have not been independently refereed or checked in a proof assistant.

2 Model, neighborhoods, and the main theorem

Let $M \geq 1$ be fixed. Let Z have a known distribution with $|Z| \leq L$ almost surely. Let $U_1, \dots, U_M$ be independent uniforms on $[-1, 1]$, and let G be standard Gaussian, independent of all other variables.

Fix $A > 0$ and set

$$\mathcal{A}_M = \{a \in [0, A]^M : a_1 \leq \dots \leq a_M\}, \quad X_{a,v} = Z + \sum_{i=1}^M a_i U_i + \sqrt{v} G, \quad P_{a,v} = \mathcal{L}(X_{a,v}). \quad (2.1)$$

There are n independent observations. The variance is either known, $v = v_0 > 0$, or unknown in a fixed compact interval $J = [v_-, v_+] \subset (0, \infty)$ with $v_- < v_+$.

Choose a reference vector

$$a^0 = (\underbrace{0, \dots, 0}_r, \underbrace{b_1, \dots, b_1}_{m_1}, \dots, \underbrace{b_s, \dots, b_s}_{m_s}), \quad 0 < b_1 < \dots < b_s < A, \quad r + \sum_{j=1}^s m_j = M. \quad (2.2)$$

The zero block may be absent, and the positive blocks may all be absent. Let $I_0, I_1, \dots, I_s$ be their consecutive index sets, omitting I_0 when $r = 0$. Choose $\rho > 0$ so small that the enlarged intervals $[0, 2\rho]$ and $[b_j - 2\rho, b_j + 2\rho]$ are disjoint and the positive ones lie inside $(0, A)$. Define

$$\Theta_\rho = \{(a, v) : a \in \mathcal{A}_M, \|a - a^0\|_\infty \leq \rho, v \in J\}. \quad (2.3)$$

In the known-variance experiment replace J by $\{v_0\}$. This is a neighborhood of a stratum, not the stratum itself: repeated half-lengths may split, and zero half-lengths may become positive.

For a block I_j , the loss is $d_j(a, \tilde{a}) = \max_{i \in I_j} |a_i - \tilde{a}_i|$. Let $\mathcal{R}_{j,n}$ be its minimax expected loss over Θ_ρ . Let $\mathcal{R}_{v,n}$ use absolute variance loss. The infimum is over all measurable estimators. Write $f_n \asymp g_n$ when their ratio is bounded above and below by positive constants for all sufficiently large n .

Theorem 2.1 (Sharp local blockwise rates). *For every fixed neighborhood above, the following rates hold:*

Quantity	Known variance	Unknown variance
Positive block I_j	$n^{-1/(2m_j)}$	$n^{-1/(2m_j)}$
Zero block I_0 , if $r \geq 1$	$n^{-1/(4r)}$	$n^{-1/(4r+4)}$
Gaussian variance	not estimated	$n^{-1/(2r+2)}$

Every displayed entry is an $\asymp$ statement for the corresponding minimax risk. One estimator, constructed on the full parameter space without knowing a^0 , r , the positive centers, or the multiplicities, attains all the upper bounds simultaneously on every fixed such neighborhood.

The constants can depend on M, A, L , the background law, the variance interval, the centers, and ρ . No uniformity is asserted as the cluster gaps or the Gaussian variance tend to zero, or as M tends to infinity.

Corollary 2.2 (Joint half-length loss). *Put $Q = \max_j m_j$, with $Q = 0$ if there is no positive block. Define*

$$D_K = \max\{Q, 2r\}, \quad (2.4)$$

$$D_U = \begin{cases} Q, & r = 0, \\ \max\{Q, 2r + 2\}, & r \geq 1. \end{cases} \quad (2.5)$$

The local minimax risk for $\|\hat{a} - a\|_\infty$ is $n^{-1/(2D_K)}$ with known variance and $n^{-1/(2D_U)}$ with unknown variance.

Proof. The maximum of finitely many block losses is bounded above by their sum and below by each individual block loss. Apply Theorem 2.1. $\square$

Remark 2.3 (The absent-block convention matters). When $r = 0$, there is no zero-block rate. Inserting $r = 0$ into the expression $n^{-1/(4r+4)}$ would introduce a nonexistent $n^{-1/4}$ obstruction. In particular, distinct strictly positive half-lengths and the unknown variance are all estimable at rate $n^{-1/2}$.

3 Cumulants turn the problem into shifted moment inversion

Set $x_i = a_i^2$ and $p_k(x) = \sum_i x_i^k$. For U uniform on $[-1, 1]$,

$$\mathbb{E}e^{tU} = \frac{\sinh t}{t}, \quad c_k := \kappa_{2k}(U) = \frac{2^{2k} B_{2k}}{2k} = (-1)^{k+1} \frac{(2k)! \zeta(2k)}{k \pi^{2k}} \neq 0. \quad (3.1)$$

Here $B_2 = 1/6$. One derivation is to logarithmically expand $\sinh t/t = \prod_{\nu \geq 1} (1 + t^2/(\pi^2 \nu^2))$ near zero. The coefficient of t^{2k} in its logarithm is $(-1)^{k+1} \zeta(2k)/(k \pi^{2k})$. In particular, $c_1 = 1/3$, $c_2 = -2/15$, and $c_3 = 16/63$.

Independence gives

$$\kappa_2(X_{a,v}) - \kappa_2(Z) = v + \frac{1}{3} p_1(x), \quad (3.2)$$

$$\kappa_{2k}(X_{a,v}) - \kappa_{2k}(Z) = c_k p_k(x), \quad k \geq 2. \quad (3.3)$$

Thus known variance gives $p_1, \dots, p_M$, while unknown variance gives

$$\eta(a, v) = \left(v + \frac{1}{3} p_1(x), p_2(x), \dots, p_{M+1}(x) \right). \quad (3.4)$$

The first power sum is hidden, not the higher ones.

The same reduction includes the Rvachev up background. It can be defined without any normalization ambiguity by

$$Z_{\text{up}} = \sum_{j=1}^{\infty} 2^{-j} U_j, \quad |Z_{\text{up}}| \leq 1, \quad \mathbb{E}e^{itZ_{\text{up}}} = \prod_{j=1}^{\infty} \text{sinc}(2^{-j}t). \quad (3.5)$$

This is the compactly supported up-law normalization discussed in the Fabius literature [6]. Its known cumulants are

$$\kappa_{2k}(Z_{\text{up}}) = \frac{c_k}{2^{2k} - 1}. \quad (3.6)$$

Only boundedness and knowledge of the background law are used below. Its smoothness, symmetry, and Fourier zeros play no role in the statistical upper bounds or the Gaussian-smoothed likelihood arguments.

4 A cluster-local shifted-power-sum theorem

We first work directly with nonnegative nodes, not their square roots. Fix disjoint ordered compact intervals $K_0, K_1, \dots, K_s$ in $[0, R]$, where K_0 starts at zero and each $K_j, j \geq 1$, is bounded away from zero. Prescribe r nodes in K_0 and m_j in K_j , with total M ; omit empty blocks. Let x, y be sorted lists obeying these same counts. For an integer $\ell \geq 0$ define

$$\Delta_{\ell}(x, y) = \max_{\ell+1 \leq k \leq \ell+M} |p_k(x) - p_k(y)|. \quad (4.1)$$

Theorem 4.1 (Localized shifted moments). *There is a constant C depending only on the fixed intervals, counts, and ℓ , such that*

$$\max_{i \in I_j} |x_i - y_i| \leq C \Delta_{\ell}(x, y)^{1/m_j}, \quad j \geq 1, \quad (4.2)$$

$$\max_{i \in I_0} |x_i - y_i| \leq C \Delta_{\ell}(x, y)^{1/(r+\ell)}, \quad r \geq 1. \quad (4.3)$$

For intervals with nonempty interiors at the indicated centers, every exponent is optimal in the corresponding local pairwise sense. In particular, omitting initial power sums changes the exponent at zero but not at a separated positive cluster.

4.1 The integer counting function

Define, outside the finitely many node locations,

$$F(t) = \sum_{i=1}^M \mathbf{1}_{\{x_i \leq t\}} - \sum_{i=1}^M \mathbf{1}_{\{y_i \leq t\}}. \quad (4.4)$$

It vanishes in the gaps between the prescribed intervals, since the counts in each preceding block agree. Integration by parts gives

$$p_k(x) - p_k(y) = -k \int_0^R t^{k-1} F(t) dt. \quad (4.5)$$

Within a block containing m nodes from each list, the nonzero signs of F can change at most $m - 1$ times. To see this, refine coincident jumps into unit jumps. A path beginning and ending at zero with m upward and m downward steps has at most m nonzero excursions. Successive opposite signs require different excursions. Ignoring zero intervals or excursions of zero length can only decrease the number of sign changes.

Choose one real root for each internal sign change, inside the corresponding zero interval or at the transition point. If necessary choose one further root at the midpoint of a prescribed interblock gap separating consecutive active blocks to match the signs in the two blocks. When inactive blocks lie between them, choose one of the genuine gaps, not a point inside an inactive block. There is a polynomial q , up to an overall sign, whose roots are these chosen points and such that

$$q(t)F(t) \geq 0 \quad \text{almost everywhere,} \quad \deg q \leq M - 1. \quad (4.6)$$

Its leading coefficient has absolute value one. At most $m_j - 1$ of its roots lie in block j , and at most $r - 1$ lie in the zero block. All roots lie in a fixed bounded interval, so the sum of the absolute values of its coefficients is bounded by a fixed constant. It follows from (4.5) that

$$\int_0^R t^\ell |q(t)| |F(t)| dt \leq C \Delta_\ell(x, y). \quad (4.7)$$

Crucially, no cancellation between blocks remains in this inequality.

4.2 A polynomial norm lower bound

Lemma 4.2. *For fixed integers $d, \ell \geq 0$ there is $c_{d,\ell} > 0$ such that every monic real polynomial Q of degree d satisfies*

$$\int_0^1 u^\ell |Q(u)| du \geq c_{d,\ell}. \quad (4.8)$$

Consequently, for any interval $[\alpha, \alpha + h] \subset [0, \infty)$ and any monic polynomial Q of degree d ,

$$\int_\alpha^{\alpha+h} t^\ell |Q(t)| dt \geq c_{d,\ell} h^{d+\ell+1}. \quad (4.9)$$

Proof. The integral defines a norm on the finite-dimensional space of polynomials of degree at most d , because its weight is positive on $(0, 1)$. The leading coefficient is a continuous linear functional in that norm. A monic polynomial therefore has norm bounded below. For the second assertion put $t = \alpha + hu$; $h^{-d}Q(\alpha + hu)$ is monic in u , and $(\alpha + hu)^\ell \geq h^\ell u^\ell$. $\square$

4.3 Proof of the inverse inequalities

Choose an index i in a block where $h = |x_i - y_i|$ attains its block maximum. If $x_i < y_i$, then $F(t) \geq 1$ throughout (x_i, y_i) : at least i entries of x and at most $i - 1$ entries of y lie to its left. The reverse ordering gives $F(t) \leq -1$. Thus $|F| \geq 1$ on an interval of length h inside that block.

Factor q into roots internal to the block and roots external to it. The latter are a fixed positive distance from the entire block: internal roots of other blocks are separated by the prescribed gaps, and roots placed in gaps are their midpoints. The external product has absolute value bounded below by a fixed positive constant. If there are $d \leq m_j - 1$ internal roots in a positive block, then t^ℓ is bounded below there and Lemma 4.2 gives

$$\int_{x_i \wedge y_i}^{x_i \vee y_i} t^\ell |q(t)| |F(t)| dt \geq ch^{d+1} \geq c'h^{m_j}. \quad (4.10)$$

The last adjustment uses only the fixed upper bound on h . In the zero block, the weighted version instead yields $c'h^{r+\ell}$. Combining with (4.7) proves (4.2)–(4.3). The sharpness constructions are given in Section 8, completing the theorem.

Corollary 4.3 (Global identifiability). *Any M consecutive positive-order power sums beginning at $\ell + 1$ determine an ordered nonnegative list of length M . On a bounded node interval the inverse has global exponent $1/(M + \ell)$.*

Proof. Regard all nodes as belonging to the single interval $K_0 = [0, R]$. □

For half-lengths, square roots are Lipschitz on a fixed positive block, while $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$ at zero. Hence the local inverse exponents for half-lengths are $1/m_j$ at a positive block, $1/(2r)$ at zero with $\ell = 0$, and $1/(2r + 2)$ at zero with $\ell = 1$. Notice that these are *moment-to-parameter* exponents; statistical rates require an additional factor $1/2$ in their powers of n .

5 A positivity certificate for the missing first power sum

A bound on individual roots is not enough to obtain the sharp variance rate. Summing the positive-root bounds would incorrectly introduce their collision multiplicities into that rate. The following argument avoids that loss.

Write $e_k(x)$ for the elementary symmetric polynomials, with $e_0 = 1$, and

$$E_x(z) = \prod_{i=1}^M (1 + x_i z) = \sum_{k=0}^M e_k(x) z^k. \quad (5.1)$$

All its coefficients are nonnegative when the nodes are nonnegative.

Theorem 5.1 (Boundary-localized recovery of p_1). *Suppose $x, y \in [0, R]^M$ each have at least $M - r$ entries greater than or equal to a fixed $b > 0$. Then*

$$|p_1(x) - p_1(y)| \leq C \left(\max_{2 \leq k \leq M+1} |p_k(x) - p_k(y)| \right)^{1/(r+1)}. \quad (5.2)$$

For $r = M$, no lower bound b is needed. On the mixed neighborhoods of Section 2, the exponent is optimal. When $r = 0$, p_1 is locally Lipschitz regardless of positive-node collisions.

Proof. Let δ denote the maximum in (5.2). By swapping x, y , assume $u = p_1(x) - p_1(y) \geq 0$. In the truncated formal power-series ring modulo z^{M+2} ,

$$E_x(z) = E_y(z) \exp(uz + R_\delta(z)), \quad R_\delta(z) = \sum_{k=2}^{M+1} \frac{(-1)^{k-1}}{k} (p_k(x) - p_k(y)) z^k. \quad (5.3)$$

This is the logarithmic generating-function identity for the e_k . On the fixed bounded node set, all coefficients of $E_y(z)e^{uz}(e^{R_\delta(z)} - 1)$ through degree $M + 1$ are $O(\delta)$. For $\delta \leq 1$, this follows directly by expanding the truncated exponential: every nonconstant term contains a coefficient of R_δ , and there are finitely many terms. For larger δ the desired conclusion follows from boundedness after increasing C .

The coefficient of z^{M+1} in E_x is zero. Therefore

$$0 \leq [z^{M+1}]E_y(z)e^{uz} = \sum_{k=0}^M e_k(y) \frac{u^{M+1-k}}{(M+1-k)!} \leq C\delta. \quad (5.4)$$

Each term is nonnegative. Keeping the term $k = M - r$ gives

$$e_{M-r}(y) \frac{u^{r+1}}{(r+1)!} \leq C\delta. \quad (5.5)$$

The coefficient $e_{M-r}(y)$ is at least b^{M-r} ; when $r = M$ it equals one. This proves the bound. Sharpness for $r \geq 1$ follows from the zero-block paths in Section 8, where the change in p_1 is of order h^2 but δ is of order h^{2r+2} . For $r = 0$, varying one positive node shows that no exponent larger than one is possible. $\square$

Example 5.2 (A cubic certificate in a mixed configuration). For the squared-width base $(0, 0, 1, 1, 3)$, $E_0(z) = (1 + z)^2(1 + 3z)$. Thus

$$[z^6]E_0(z)e^{uz} = \frac{u^3}{2} + \frac{7u^4}{24} + \frac{u^5}{24} + \frac{u^6}{720}. \quad (5.6)$$

The first nonzero power is u^3 because there are two zero nodes. The repeated positive node at 1 does not change that order. This is the algebra behind the variance exponent $1/(r+1)$.

6 Regular variance estimation at positive collisions

There is more than a Lipschitz bound when all half-lengths are positive. Although recovering separate roots is singular at a positive collision, recovering their sum is an ordinary implicit-function problem.

Theorem 6.1 (Analytic recovery of the coefficient data). *At any nonnegative node vector x with $\prod_i x_i > 0$, the first power sum p_1 and every $e_j(x)$ are real-analytic functions of $(p_2, \dots, p_{M+1})$ on a neighborhood of that moment vector. The analytic extension is defined on an open set in moment space; it need not send every perturbed vector to a polynomial with only real roots. Its differential is*

$$dp_1 = \sum_{k=2}^{M+1} \frac{(-1)^k e_{M+1-k}(x)}{k e_M(x)} dp_k. \quad (6.1)$$

Proof. Define e_{M+1} as the coefficient of z^{M+1} in

$$\exp \left(\sum_{k=1}^{M+1} \frac{(-1)^{k-1}}{k} p_k z^k \right). \quad (6.2)$$

This coefficient is a polynomial in the p_k and vanishes for M actual nodes. Differentiating its generating function gives

$$\frac{\partial e_{M+1}}{\partial p_k} = \frac{(-1)^{k-1}}{k} e_{M+1-k}. \quad (6.3)$$

In particular, the derivative with respect to p_1 is $e_M = \prod_i x_i > 0$. The real-analytic implicit-function theorem supplies p_1 as a local function of the other moments. Implicit differentiation yields (6.1), and Newton identities give each e_j . Distinct roots were never required. $\square$

Here is an explicit statistical consequence. Put $M_X(t) = \mathbb{E}e^{tX}$, where X has the distribution at the parameter under consideration. Define the centered cumulant influence polynomials

$$\psi_j(y) = j![t^j] \left(\frac{e^{ty}}{M_X(t)} - 1 \right), \quad j \geq 1. \quad (6.4)$$

Only a finite Taylor expansion is used. Each ψ_j is a monic polynomial of degree j and has expectation zero. For example, $\psi_2(y) = (y - \mathbb{E}X)^2 - \text{Var}(X)$.

Theorem 6.2 (Asymptotic normality of the variance estimator). *Suppose every $a_i > 0$ and v is an interior point of J . There is a consistent plug-in estimator $\hat{v}_{\text{an}}$ satisfying*

$$\sqrt{n}(\hat{v}_{\text{an}} - v) \implies N(0, V_v), \quad 0 < V_v = \mathbb{E}\psi_v(X)^2 < \infty, \quad (6.5)$$

where

$$\psi_v(y) = \psi_2(y) - \frac{1}{3e_M(x)} \sum_{k=2}^{M+1} \frac{(-1)^k e_{M+1-k}(x)}{kc_k} \psi_{2k}(y). \quad (6.6)$$

This remains valid when any subset of the positive half-lengths coincides. No efficiency claim is made for this particular influence polynomial.

Proof. Estimate raw moments through degree $2M + 2$ by sample averages and form their cumulant polynomials. The joint central limit theorem applies because all moments of X are finite. A first-order perturbation of $\log M_X(t)$ under an empirical-law perturbation is $e^{ty}/M_X(t) - 1$, so the influence polynomial of the empirical cumulant of order j is (6.4).

Use Theorem 6.1 to recover a real $\hat{p}_1$ from the estimated $p_2, \dots, p_{M+1}$, and set $\hat{v}_{\text{an}} = \hat{\eta}_0 - \hat{p}_1/3$. The local branch can be selected using any preliminary consistent full-model moment estimator, such as the one in the next section: choose the root of $e_{M+1} = 0$ closest to its preliminary p_1 . With probability tending to one, the correct simple root is the unique nearest one in a sufficiently small neighborhood. Arbitrary bounded values can be used outside this event, and clipping to J does not affect the limit at an interior v .

The multivariate delta method, applied to (6.1), gives (6.6). Its second moment is finite. Its coefficient of y^{2M+2} is nonzero, because the $k = M + 1$ summand has coefficient $(-1)^{M+1}/((M+1)c_{M+1})$ and $e_M > 0$. No lower-degree summand can cancel it. Gaussian convolution gives a density strictly positive everywhere, so this nonzero polynomial has positive variance. This proves (6.5). $\square$

For two identical positive squared widths $x_1 = x_2 = c > 0$, the differential is particularly transparent:

$$dp_1 = \frac{1}{c} dp_2 - \frac{1}{3c^2} dp_3, \quad dv = d\eta_0 - \frac{1}{3c} dp_2 + \frac{1}{9c^2} dp_3. \quad (6.7)$$

No inverse factor involving $|x_1 - x_2|$ occurs. In contrast, distinguishing the two half-lengths has neighborhood minimax rate $n^{-1/4}$.

7 One moment estimator adapts to every fixed stratum

7.1 Uniformly estimating a finite cumulant vector

Let $K = M$ in the known-variance case and $K = M + 1$ otherwise. Estimate raw moments through degree $2K$. The model has uniformly bounded moments of every fixed order: its non-Gaussian part is bounded by $L + MA$, and its Gaussian variance lies in a fixed compact interval. Consequently, for each $1 \leq j \leq 2K$,

$$\mathbb{E}|\hat{\mu}_j - \mu_j| \leq C_j n^{-1/2}. \quad (7.1)$$

To obtain a uniform bound after applying cumulant polynomials, clip each empirical raw moment to a fixed compact interval containing every possible population value. Clipping cannot increase its error. On this compact box, the cumulant polynomials are Lipschitz. They can be computed recursively from

$$\kappa_j = \mu_j - \sum_{q=1}^{j-1} \binom{j-1}{q-1} \kappa_q \mu_{j-q}. \quad (7.2)$$

After subtracting the known background cumulants and dividing by the fixed nonzero c_k , the estimated observation-coordinate vector satisfies

$$\sup_{a,v} \mathbb{E} \|\hat{\eta} - \eta(a, v)\|_\infty \leq Cn^{-1/2}. \quad (7.3)$$

With known variance use $\eta = (p_1, \dots, p_M)$.

7.2 Feasible fitting without knowing the local pattern

On a deterministic grid of the entire compact parameter space, with mesh at most $1/n$ in each half-length and in v , minimize

$$\|\eta(a, v) - \hat{\eta}\|_\infty. \quad (7.4)$$

Include the endpoints and sort half-length coordinates. Break ties lexicographically. This gives a measurable estimator $(\hat{a}, \hat{v})$ without any cluster information. Since η is uniformly Lipschitz as a function of (a, v) , there is a grid point within $O(1/n)$ in coordinate error of the truth. Thus, with

$$\varepsilon = 2 \|\hat{\eta} - \eta(a, v)\|_\infty + C/n, \quad (7.5)$$

we have

$$\|\eta(\hat{a}, \hat{v}) - \eta(a, v)\|_\infty \leq \varepsilon, \quad \mathbb{E}\varepsilon \leq Cn^{-1/2}. \quad (7.6)$$

The grid proves statistical attainability; it is not advertised as an efficient large-capacity algorithm.

7.3 Localization and blockwise error

Corollary 4.3, followed by square roots, gives a uniform global continuity modulus. Choose a fixed $\varepsilon_0 > 0$ so small that $\varepsilon \leq \varepsilon_0$ forces $\|\hat{a} - a\|_\infty < \rho$. For true a in the ρ -neighborhood, $\hat{a}$ therefore has the correct numbers of roots in the enlarged 2ρ -blocks. Their squared intervals are disjoint, with the positive ones bounded away from zero.

Apply Theorem 4.1 to the true and fitted squared widths. On this event,

$$d_j(\hat{a}, a) \leq C\varepsilon^{1/m_j}, \quad j \geq 1, \quad (7.7)$$

$$d_0(\hat{a}, a) \leq \begin{cases} C\varepsilon^{1/(2r)}, & \text{known variance,} \\ C\varepsilon^{1/(2r+2)}, & \text{unknown variance,} \end{cases} \quad r \geq 1. \quad (7.8)$$

In the unknown-variance case, Theorem 5.1 and the first coordinate of (3.4) give

$$|\hat{v} - v| \leq \varepsilon + \frac{1}{3} |p_1(\hat{a}^2) - p_1(a^2)| \leq C\varepsilon^{1/(r+1)}. \quad (7.9)$$

This step is the reason positive multiplicities do not contaminate the variance rate.

For $0 < \alpha \leq 1$, Jensen's inequality gives $\mathbb{E}\varepsilon^\alpha \leq (\mathbb{E}\varepsilon)^\alpha$. On the complementary event all losses are bounded, and

$$\mathbb{P}\{\varepsilon > \varepsilon_0\} \leq \varepsilon_0^{-1} \mathbb{E}\varepsilon = O(n^{-1/2}). \quad (7.10)$$

Combining these facts with (7.8)–(7.9) proves all upper bounds in Theorem 2.1, including its adaptivity assertion. The preliminary estimator used in Theorem 6.2 is also consistent by the same global continuity argument.

Remark 7.1 (What adaptivity means here). The estimator does not require the centers, multiplicities, or zero count as inputs. Its rate on each fixed neighborhood is the corresponding sharp rate. This does not assert uniform bounds when the neighborhoods, cluster gaps, or multiplicities themselves vary with n . Such moving-stratum transitions are among the further research questions.

8 Exact moment-preserving paths and sharpness

8.1 A general shifted-moment flow at zero

Fix r distinct positive nodes $z_1 < \dots < z_r$ and let $P_z(t) = \prod_{i=1}^r (t - z_i)$. For an integer $\ell \geq 0$, consider the local ordinary differential equation

$$\dot{z}_i = \frac{1}{z_i^\ell P'_z(z_i)}. \quad (8.1)$$

The vector field is analytic while the nodes remain distinct and positive, so it has a local solution with those properties. Lagrange interpolation of monomials gives

$$\sum_{i=1}^r \frac{z_i^j}{P'_z(z_i)} = \begin{cases} 0, & 0 \leq j \leq r-2, \\ 1, & j = r-1. \end{cases} \quad (8.2)$$

Indeed, these are the coefficients of t^{r-1} in the interpolation formula for t^j on the r nodes.

Lemma 8.1 (Exact shifted cancellation). *Along (8.1),*

$$\frac{d}{dt} p_k(z) = 0 \quad (\ell + 1 \leq k \leq \ell + r - 1), \quad \frac{d}{dt} p_{\ell+r}(z) = \ell + r. \quad (8.3)$$

When $\ell = 1$,

$$\frac{d}{dt} p_1(z) = \frac{(-1)^{r+1}}{\prod_i z_i} \neq 0. \quad (8.4)$$

Proof. Differentiating a power sum gives $k \sum_i z_i^{k-\ell-1} / P'_z(z_i)$, so (8.3) follows from (8.2). For (8.4), use $1/P_z(t) = \sum_i 1/(P'_z(z_i)(t - z_i))$ and evaluate at $t = 0$. $\square$

Choose two distinct nearby times and call the resulting node lists α, β . Their power sums from $\ell + 1$ through $\ell + r - 1$ agree, while $p_{\ell+r}$ differs. Insert the lists $h^2 \alpha$ and $h^2 \beta$ into the squared-width zero block, keeping all positive blocks unchanged. Then

$$d_0(a^{(\alpha)}(h), a^{(\beta)}(h)) \asymp h, \quad \Delta_\ell((a^{(\alpha)}(h))^2, (a^{(\beta)}(h))^2) \asymp h^{2r+2\ell}. \quad (8.5)$$

There is an actual differing node, and all tuples are fixed before h tends to zero. This proves the sharp half-length zero-block exponent. Using h in place of h^2 as the node scale proves sharpness of Theorem 4.1 directly. When $\ell = 1$, the nonzero change in $p_1(\alpha) - p_1(\beta)$ also proves sharpness of Theorem 5.1.

8.2 A positive collision of arbitrary multiplicity

Fix a positive squared-width center $c = b_j^2$ and put $q = m_j$. Choose a monic polynomial $P(t) = \prod_{i=1}^q (t - u_i)$ with distinct real roots. For a small nonzero constant τ , the polynomial $P(t) + \tau$ has distinct real roots $w_1, \dots, w_q$ close to the u_i : take small disjoint intervals around the simple roots on which the endpoint signs persist under the perturbation. Newton identities imply

$$p_k(u) = p_k(w) \quad (1 \leq k < q), \quad p_q(u) - p_q(w) = q\tau \neq 0. \quad (8.6)$$

For $q = 1$, the first set of conditions is empty.

Use squared widths $c + hu_i$ and $c + hw_i$ in block j . They are positive and belong to the required block for all sufficiently small $h > 0$. Their ordered half-length distance is of order h . For any fixed integer $k \geq 1$,

$$\sum_i (c + hu_i)^k - \sum_i (c + hw_i)^k = \sum_{d=q}^k \binom{k}{d} c^{k-d} h^d (p_d(u) - p_d(w)), \quad (8.7)$$

where an empty sum is zero. Thus every finite collection of shifted moments differs by $O(h^q)$, and at least one moment in $\ell + 1, \dots, \ell + M$ has a nonzero coefficient of h^q . This proves the positive-block sharpness in Theorem 4.1.

These paths establish algebraic sharpness but do not, by themselves, establish a sampling lower bound. That additional step is the purpose of the next section.

9 Weighted likelihood expansions

We use the conventions

$$H^2(P, Q) = \int (\sqrt{p} - \sqrt{q})^2 dy, \quad \chi^2(P, Q) = \int \frac{(p - q)^2}{q} dy. \quad (9.1)$$

The Gaussian component ensures that every density involved is strictly positive.

9.1 A common domination argument

Lemma 9.1 (From density jets to likelihood distance). *Suppose two one-parameter families arise by convolving Gaussian densities with uniformly bounded random shifts that depend smoothly on h , and by allowing their Gaussian variances to depend smoothly on h . Assume their variances stay in a sufficiently small interval $[v_*, v^*]$ with $v^* < 2v_*$, and the shift derivatives needed below are uniformly bounded. Suppose $p_0 = q_0 = f_0$ and the Taylor expansions satisfy*

$$p_h(y) - q_h(y) = h^d g(y) + O(h^{d+1} W(y)), \quad (9.2)$$

where the remainder has a common Gaussian-times-polynomial envelope. Then

$$\chi^2(P_h, Q_h) = h^{2d} \int \frac{g^2}{f_0} dy + o(h^{2d}), \quad (9.3)$$

$$H^2(P_h, Q_h) = \frac{h^{2d}}{4} \int \frac{g^2}{f_0} dy + o(h^{2d}). \quad (9.4)$$

The integral is finite and is positive when g is not identically zero.

Proof. Let B bound all shifts. Chain-rule differentiation of a Gaussian, including variance derivatives, gives for each required derivative an upper bound

$$W(y) = C(1 + |y|)^D \exp\left(-\frac{((|y| - B)_+)^2}{2v^*}\right). \quad (9.5)$$

Expectations preserve the bound. Each density has a lower bound

$$p_h(y), q_h(y) \geq c \exp\left(-\frac{(|y| + B)^2}{2v_*}\right) =: \underline{f}(y). \quad (9.6)$$

The prefactors can be chosen uniformly on the variance interval. The ratio $W(y)^2/\underline{f}(y)$ is integrable because its quadratic exponent has leading coefficient $-1/v^* + 1/(2v_*) < 0$. Taylor's theorem therefore gives convergence of $h^{-d}(p_h - q_h)$ to g with an integrable squared envelope after division by either density. Dominated convergence proves (9.3). For Hellinger distance use

$$(\sqrt{p} - \sqrt{q})^2 = \frac{(p - q)^2}{(\sqrt{p} + \sqrt{q})^2}, \quad (9.7)$$

whose denominator tends pointwise to $4f_0$ and is bounded below by a constant times $\underline{f}$. Positivity of the integral follows from $f_0 > 0$ and $g \neq 0$. $\square$

The variance interval condition is local to the hard paths. Even when the original interval J is large, choose an interior v_0 and take h small enough for both paths to lie in such a small interval. It is not an additional restriction on the model in Theorem 2.1.

9.2 Positive-block likelihood separation

Use the positive paths (8.6), keep $v = v_0$ fixed, and keep every other block fixed at its reference value. Regard the density as a smooth symmetric function of the q squared widths near $(c, \dots, c)$. This smoothness follows from differentiating

$$\mathbb{E} \phi_{v_0} \left(y - Z - \sum_i \sqrt{x_i} U_i \right), \quad (9.8)$$

with all varied x_i bounded away from zero. Its homogeneous Taylor coefficient of degree d in the offsets is a symmetric homogeneous polynomial in $u_1, \dots, u_q$, with coefficient functions of y . Every such polynomial is a polynomial in $p_1(u), \dots, p_d(u)$: first use elementary symmetric polynomials and then Newton identities, noting the degrees. Hence (8.6) cancels all density Taylor coefficients below degree q .

There is therefore a function $g_{q,c}$ such that

$$p_h - q_h = h^q g_{q,c} + O(h^{q+1}W). \quad (9.9)$$

It is not zero. Indeed, the cumulant of order $2q$ differs by

$$\kappa_{2q}(P_h) - \kappa_{2q}(Q_h) = c_q h^q (p_q(u) - p_q(w)) \neq 0. \quad (9.10)$$

For $q = 1$ the same formula uses the second cumulant and the fixed Gaussian variance. If $g_{q,c}$ vanished, the envelope would force all fixed raw-moment differences, and therefore this cumulant difference, to be $O(h^{q+1})$, a contradiction. Lemma 9.1 yields

$$H^2(P_h, Q_h) \asymp \chi^2(P_h, Q_h) \asymp h^{2q}. \quad (9.11)$$

9.3 Zero-block likelihood separation with known variance

Let f_0 be the density of the common background, all the fixed positive uniform factors, and a Gaussian of variance v_0 . Choose α, β from the $\ell = 0$ flow. For the r small half-lengths use $h\sqrt{\alpha_i}$ and $h\sqrt{\beta_i}$, respectively. Since their first $r - 1$ power sums agree, their logarithmic moment generating functions first differ at order h^{2r} . Finite Taylor expansion of the Gaussian under the bounded shifts gives

$$p_h - q_h = h^{2r} \gamma_r f_0^{(2r)} + O(h^{2r+1}W), \quad \gamma_r = \frac{c_r (p_r(\alpha) - p_r(\beta))}{(2r)!} \neq 0. \quad (9.12)$$

For clarity, this is a statement about finite Taylor coefficients, not an assumption that an infinite differential-operator series converges. At the finite-jet level, translation contributes e^{-hWD} and changing

Gaussian variance by dh^2 contributes $e^{dh^2 D^2/2}$, where $D = d/dy$. Averaging their finite products reproduces the cumulant expansion; all relevant derivative orders here are even.

The function $f_0^{(2r)}$ is not identically zero. Otherwise f_0 would be a polynomial of degree at most $2r - 1$, incompatible with being a nonzero integrable density. Lemma 9.1 now gives

$$H^2(P_h, Q_h) \sim \frac{\gamma_r^2 h^{4r}}{4} \int \frac{(f_0^{(2r)})^2}{f_0} dy. \quad (9.13)$$

The corresponding chi-square equivalent is four times as large.

9.4 Exact Gaussian compensation at zero

Choose α, β from the $\ell = 1$ flow. The small uniform widths remain $h\sqrt{\alpha_i}$ and $h\sqrt{\beta_i}$, but set

$$v_\alpha(h) = v_0 - \frac{h^2}{3} p_1(\alpha), \quad v_\beta(h) = v_0 - \frac{h^2}{3} p_1(\beta). \quad (9.14)$$

Both lie in J for small h if v_0 is interior. Each total second cumulant is exactly that of f_0 . The higher cumulants of orders $4, \dots, 2r$ also agree, by the flow. Consequently,

$$p_h - q_h = h^{2r+2} \gamma_{r+1} f_0^{(2r+2)} + O(h^{2r+3} W), \quad \gamma_{r+1} = \frac{c_{r+1}(p_{r+1}(\alpha) - p_{r+1}(\beta))}{(2r+2)!} \neq 0. \quad (9.15)$$

The same domination argument proves

$$H^2(P_h, Q_h) \sim \frac{\gamma_{r+1}^2 h^{4r+4}}{4} \int \frac{(f_0^{(2r+2)})^2}{f_0} dy, \quad \chi^2(P_h, Q_h) \asymp h^{4r+4}. \quad (9.16)$$

Meanwhile the variance difference is of order h^2 , by (8.4). This identifies both the scale-estimation obstruction and the sharper variance-estimation obstruction in the same pair of experiments.

10 Matching minimax lower bounds

For completeness, a two-point argument suffices; no mixture-identifiability theorem is imported. If two parameters are separated by d in the chosen metric, any estimator induces a test by choosing whichever of the two parameters is closer. The triangle inequality implies

$$\max_{i=0,1} \mathbb{E}_i[\text{loss}] \geq \frac{d}{4} (1 - d_{\text{TV}}(P_0^{\otimes n}, P_1^{\otimes n})). \quad (10.1)$$

Moreover,

$$1 + \chi^2(P^{\otimes n}, Q^{\otimes n}) = (1 + \chi^2(P, Q))^n, \quad d_{\text{TV}}(P, Q) \leq \frac{1}{2} \sqrt{\chi^2(P, Q)}. \quad (10.2)$$

Thus $\chi^2(P, Q) \leq c/n$ with sufficiently small fixed c makes the final factor in (10.1) bounded below by a positive constant.

For positive block j , use (9.11) and choose $h = c_0 n^{-1/(2m_j)}$. The half-length separation is of order h and the χ^2 distance is $O(1/n)$. These alternatives use the same variance, so the lower bound applies in both experiments.

For the zero block with known variance, use (9.13) with $h = c_0 n^{-1/(4r)}$. With unknown variance, use (9.16) with $h = c_0 n^{-1/(4r+4)}$. Both have half-length separation of order h . In the latter construction, the variance separation is of order $h^2 = n^{-1/(2r+2)}$, proving the variance lower bound for $r \geq 1$.

If $r = 0$, hold every half-length fixed and vary an interior variance by an amount t . Differentiation with respect to variance gives the density expansion $f_{v_0+t} - f_{v_0} = t f''_{v_0}/2 + O(t^2 W)$, so its chi-square distance is $O(t^2)$. Taking $t = c_0 n^{-1/2}$ proves the remaining variance lower bound.

All constructed alternatives lie in Θ_ρ for large enough n and approach the same reference factorization. This completes the proof of Theorem 2.1.

Remark 10.1 (Two moving alternatives are essential). The hard pairs generally have *both* endpoints depending on n . The theorem concerns minimax risk on a fixed neighborhood, not a testing problem with one endpoint held equal to a^0 , and not the risk at a singleton parameter set. High-order cancellation between two splitting configurations can be much stronger than the cancellation between either configuration and the unsplit reference. Conflating these questions can produce incorrect “local” exponents.

11 Examples, exact constants, and reproducible checks

11.1 Three mixed configurations

Consider the reference half-length vector

$$a^0 = (0, 0, 1, 1, 1, 2, 3), \quad M = 7, \quad (11.1)$$

with $A > 3$. There are two zero slots, a positive triple collision, and two simple positive factors. On a sufficiently small fixed neighborhood, Theorem 2.1 gives the following simultaneous rates.

Target	Known variance	Unknown variance
The two small factors	$n^{-1/8}$	$n^{-1/12}$
The three factors near 1	$n^{-1/6}$	$n^{-1/6}$
Each factor near 2 or 3	$n^{-1/2}$	$n^{-1/2}$
Gaussian variance	not estimated	$n^{-1/6}$
All half-lengths jointly	$n^{-1/8}$	$n^{-1/12}$

The earlier global unknown-variance rate at capacity seven is $n^{-1/32}$ [1]. It is attained near the all-zero configuration, not uniformly as the local obstruction at every factorization. The mixed rate $n^{-1/12}$ is faster, while its simple positive factors are faster still.

For a second example, suppose there are five identical positive factors and one different positive factor, with no zero slots. The five-factor block has rate $n^{-1/10}$, but the remaining factor and the Gaussian variance both have rate $n^{-1/2}$. The analytic variance estimator of Theorem 6.2 applies directly at the fivefold collision.

For a third example, take one zero slot and a positive block of multiplicity six. The joint half-length rate is $n^{-1/12}$ in both variance experiments: the positive block is the slowest component. Nevertheless, the zero-block rate itself changes from $n^{-1/4}$ to $n^{-1/8}$ when variance becomes unknown, and the variance has rate $n^{-1/4}$. A single joint exponent can therefore hide a genuine loss of information in one part of the model.

11.2 A fully explicit unknown-variance hard pair

Take two zero slots, no background, and limiting Gaussian variance 1. Use the fixed squared-scale lists

$$\alpha = (1, 8), \quad \beta = (4, 7). \quad (11.2)$$

Their power sums obey

$$p_1(\alpha) - p_1(\beta) = -2, \quad p_2(\alpha) = p_2(\beta) = 65, \quad p_3(\alpha) - p_3(\beta) = 106. \quad (11.3)$$

Define the half-length and variance pairs

$$a_\alpha(h) = (h, \sqrt{8}h), \quad v_\alpha(h) = 1 - 3h^2, \quad (11.4)$$

$$a_\beta(h) = (2h, \sqrt{7}h), \quad v_\beta(h) = 1 - \frac{11}{3}h^2. \quad (11.5)$$

The ordered half-length distance is exactly h , and the variance distance is $2h^2/3$. Both total variances equal one. Let ϕ be the standard Gaussian density. Since $c_3 = 16/63$,

$$p_h - q_h = \frac{106}{2835} h^6 \phi^{(6)} + O(h^7 W). \quad (11.6)$$

For every nonnegative integer d ,

$$\int \frac{(\phi^{(d)})^2}{\phi} dy = d!. \quad (11.7)$$

One direct proof writes $\phi^{(d)}/\phi = (-1)^d \text{He}_d$ and uses

$$\mathbb{E} \left[e^{tG - t^2/2} e^{sG - s^2/2} \right] = e^{ts}.$$

Equating the coefficients of $t^d s^d$ proves (11.7); this uses only finite coefficient identities. Therefore the exact likelihood constants are

$$\lim_{h \downarrow 0} \frac{\chi^2(P_h, Q_h)}{h^{12}} = 6! \left(\frac{106}{2835} \right)^2 = \frac{179776}{178605}, \quad (11.8)$$

$$\lim_{h \downarrow 0} \frac{H^2(P_h, Q_h)}{h^{12}} = \frac{44944}{178605}. \quad (11.9)$$

This one example witnesses both the $n^{-1/12}$ half-length obstruction and the $n^{-1/6}$ variance obstruction for a two-dimensional zero block.

For comparison, with known variance take $\alpha = (1, 4)$ and $\beta = (2, 3)$. Their first power sums agree and their second power sums differ by 4. The leading density difference is $-h^4 \phi^{(4)}/45$, giving

$$\lim_{h \downarrow 0} \frac{\chi^2(P_h, Q_h)}{h^8} = \frac{8}{675}, \quad \lim_{h \downarrow 0} \frac{H^2(P_h, Q_h)}{h^8} = \frac{2}{675}. \quad (11.10)$$

These are exact algebraic consequences of the likelihood lemma, not fitted exponents from numerical data.

11.3 Verification program and numerical diagnostic

The accompanying `verify_results.py` uses exact rational and symbolic arithmetic for 186 assertions. It checks the shifted-flow tangent identities for $1 \leq r \leq 8$ and $0 \leq \ell \leq 3$, the additional first-sum identity for $\ell = 1$, Newton-sum cancellation under constant-coefficient perturbations through multiplicity seven, the two explicit hard pairs, and the mixed positivity certificate (5.6).

An optional high-precision diagnostic evaluates the unknown-variance pair (11.5). The density of a Gaussian convolved with uniforms of half-lengths $a, b > 0$ can be evaluated as

$$f(y; a, b, v) = \frac{T_v(y + a + b) - T_v(y + a - b) - T_v(y - a + b) + T_v(y - a - b)}{4ab}, \quad (11.11)$$

where

$$T_v(t) = t\Phi(t/\sqrt{v}) + \sqrt{v}\phi(t/\sqrt{v}). \quad (11.12)$$

Indeed, $T_v'' = \phi_v$, so two integrations of the Gaussian density over the uniform intervals give (11.11). The recorded 65-digit computation integrates over $[-12, 12]$ and gives:

h	χ^2/h^{12}	H^2/h^{12}
0.12	1.029200612837	0.257277110688
0.09	1.014079389051	0.253515089592
0.06	1.008104404930	0.252025629178
0.04	1.006868334558	0.251717039656
Exact limiting value	1.006556367403	0.251639091851

The finite integration window and floating-point arithmetic mean that this is corroboration, not an interval certificate. In particular, the displayed decimal places are numerical outputs, not certified error bounds. The exact limits follow from the proofs, independently of this table. No simulation of an estimator’s minimax risk is claimed.

12 Implementation and formalization boundaries

12.1 A concrete inference pipeline

The statistical construction can be implemented without differentiating an estimated characteristic function. Compute sample raw moments, convert them to cumulants using (7.2), subtract the known background cumulants, and fit the feasible coordinate vector (3.4). For a fixed small capacity, exhaustive grids with deterministic tie-breaking are a complete implementation of the estimator used in the proof. More efficient nonlinear optimization is a useful replacement only if its residual error is controlled. An approximate optimizer whose moment residual exceeds the true optimum by $O(n^{-1/2})$ preserves the proved rates: simply add that error to ε in (7.6).

When the preliminary estimate lies near a configuration with all widths positive, variance can instead be estimated by the polynomial branch described in Section 6. Its degree in the unknown p_1 is $M + 1$, and the branch-selection step is essential. Solving the equation without identifying the branch can produce an algebraically valid but statistically irrelevant answer. Positivity and real-rootedness should be enforced when individual widths, rather than only the analytic variance functional, are recovered.

The supplied program verifies algebra and evaluates a likelihood diagnostic; it is not a general-purpose implementation of this inference pipeline. The existence and rates of the grid estimator are proved in the article. Their proof does not establish polynomial computational complexity in M .

12.2 A modular path toward proof-assistant verification

A useful formalization should separate four logically different components. First, finite symmetric-polynomial identities cover cumulants, the generating function for e_k , the coefficient positivity certificate, and the Lagrange sum identities. These are exact algebraic statements and are natural early targets. Second, the ordered counting-function argument requires a finite sign-change lemma, a sign-matching polynomial, integration by parts for step functions, and a polynomial norm estimate. Those are the essential analytic ingredients of the new inverse theorem.

Third, the stochastic layer requires finite-moment bounds, cumulant estimation, measurable grid minimization, and the elementary two-point testing bound. Fourth, the likelihood layer requires differentiating Gaussian mixtures under bounded shifts and dominated convergence with explicit Gaussian upper and lower envelopes. Replacing that last layer by unweighted Taylor expansions would lose the statistical lower-bound proof.

This article does not supply purportedly checked Lean or Rocq code. Its symbolic program checks finite instances and identities, not all real-node configurations or all sample sizes. The claim ledger records which statements have complete conventional proofs, which have finite checks, and which remain research questions. No external theorem about a finite *mixture* is silently used to certify this different convolution model.

13 Further research questions

The following are proposed extensions, not conclusions already proved here. The fixed-neighborhood classification provides both candidate rates and specific obstacles for them.

Research question 1 (Moving gaps and multiscale transitions). Determine sharp risk bounds when positive cluster gaps and positive centers approach zero at prescribed rates depending on n .

The constants hidden in Theorem 4.1 contain separation factors from roots outside the target block. Tracking those factors explicitly should identify when two positive blocks behave separately, when they behave as one larger collision, and when a positive collision crosses into the zero-boundary regime. A complete answer needs matching transition inequalities, not just substitution of an n -dependent gap into one nonuniform bound. The simplest nontrivial test case is two small, nearly coincident positive factors with unknown Gaussian variance.

Research question 2 (Limit experiments at singular strata). Identify the joint local limit experiments and optimal rescaled estimators at configurations with both zero and positive collision blocks.

The proof identifies first nonzero likelihood jets but does not describe the full set of attainable jets under all local perturbations. That set is constrained by real-rootedness and nonnegative squared widths. A promising formulation uses regular moment coordinates and their feasible tangent cones, followed by a singular inverse map to widths. It should explain the joint behavior of a root- n variance functional and slower splitting coordinates, and clarify which limits are non-Gaussian. Ordinary unconstrained local asymptotic normality should not be assumed at the zero boundary.

Research question 3 (Efficient variance inference at positive collisions). Determine the exact asymptotic information bound for Gaussian variance when all uniform widths are positive, including repeated widths.

The explicit polynomial influence function (6.6) proves attainability and asymptotic normality, but need not minimize asymptotic variance among all estimators using the full sample. The relevant nuisance geometry includes splitting directions that appear at higher order in the raw width parameterization. A full solution should identify the appropriate regular functional and nuisance tangent space, compare its efficient influence function with the polynomial one, and construct an estimator attaining the bound without requiring distinct roots.

Research question 4 (Honest adaptive confidence regions). Can one construct uniformly valid confidence regions whose blockwise widths adapt to the local exponents without prior knowledge of the strata?

Moment confidence sets followed by feasible inversion are a concrete starting point. Point estimation adapts by Section 7, but confidence-set coverage near changing multiplicities and near zero is a stronger demand. An informative result would specify finite-sample or uniform asymptotic coverage, prove diameter bounds on fixed separated neighborhoods, and describe necessary losses at the transition scales. Exact multiplicity selection should not be substituted for uncertainty quantification when the data cannot reliably distinguish a collision from a small split.

Research question 5 (Mixed strata without a positive variance floor). Extend the blockwise classification to zero Gaussian variance and to variances that decrease with n , for Fabius–Rvachev and more general flat backgrounds.

The repository’s flat-boundary report already supplies global zero-noise results for infinite uniform backgrounds [2]; those global results should be treated as the starting point, not reasserted as an open problem. The missing issue is the mixed-stratum statistical geometry and its possible uniformity as variance vanishes. The present algebraic upper bounds survive, but the Gaussian lower-envelope argument does not. One must replace it with weighted square-root expansions for the relevant positive-collision paths, while tracking any changes caused by finite-order support boundaries.

Research question 6 (Certified, computationally efficient moment fitting). Find a practical solver with a proved residual certificate and controlled complexity in both capacity and target accuracy.

The equal-multiplicity power-sum image is more restrictive than the usual moment cone with arbitrary positive weights. This may help or obstruct convex relaxations. A useful algorithm would combine moment denoising, coefficient recovery, certified real-root isolation, and branch selection; it would either return a feasible tuple with a certified near-optimal residual or report a quantified failure. The statistical theorem immediately transfers to such an algorithm once its excess residual is bounded at the sampling noise scale. The required numerical stability near high multiplicities is part of the problem.

Research question 7 (Unknown capacity and calibrated backgrounds). What is recoverable when the capacity is not fixed in advance, or the background law is known only through an independent calibration sample?

Zero padding makes the number of genuinely nonzero factors unidentifiable without a separation requirement. A rigorous treatment should first specify minimum detectable widths or a loss that tolerates unresolved small factors. With a calibration sample for Z , subtraction of background cumulants incurs an additional error; for fixed capacity and suitable moment bounds it is natural to investigate rates driven by the combined moment-estimation error. The more difficult issue is uniformity as the capacity grows and the number of required moments increases. None of those growing-dimensional conclusions is supplied by the present fixed-capacity proof.

Research question 8 (Multidimensional uniform directions). Develop an analogue for random vectors of the form $Z + \sum_i U_i a_i + G_\Sigma$, with unknown vectors a_i and Gaussian covariance Σ .

The even cumulant tensors contain sums of the rank-one tensors $a_i^{\otimes 2k}$. Recovering the directions has an unavoidable sign and permutation ambiguity, and new degeneracies arise from parallel directions and from vanishing vectors. The one-dimensional result describes restrictions to a line but does not settle joint tensor identifiability or sharp rates. A first target is a two-direction model in the plane, distinguishing a near-parallel collision from two directions with a fixed angular separation. The covariance-confounding phenomenon then becomes matrix-valued.

14 Conclusion

The local statistical obstruction in uniform-factor recovery is not the total capacity alone. It is distributed among positive clusters and the zero block. Positive collisions contribute their multiplicity exponents; zeros contribute an additional square-root loss; and unknown Gaussian variance removes one more moment order only at that boundary. A sign-matching polynomial localizes the root-recovery problem, while a nonnegative coefficient identity protects the first power sum from positive-collision instability.

Together with weighted likelihood expansions, these facts give a complete fixed-neighborhood blockwise minimax classification under positive Gaussian smoothing. They also show that a variance parameter can remain analytically regular and asymptotically normal while individual positive factors are singular. The open directions above concern changes in the experiment or its scaling, rather than gaps in the stated fixed-capacity positive-variance classification.

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